

ActInf GuestStream #004.1 ~ Elliot Murphy

GuestStream | Elliot Murphy | 2:01:34

hello
and welcome to the active inference lab
today
is april 14th 2021 and we're here in
guest stream
number 4.1 with elliot murphy so this is
going to be a really
interesting talk and elliot we
appreciate you coming on
so i'll just pass it to you to introduce
yourself
introduce the topic and then share some
slides so anyone who's watching live
or i guess later in the comments just
ask questions and we'll have time during
this presentation and at the end
to go over some comments but thanks
again elliot and looking forward to
hearing
yeah thank you very much and so i am a
postdoc working at ut health
uh in houston texas i work on
intracranial recordings of
epilepsy patients doing a bunch of
language tasks trying to explore the
neural basis of language
and today i will i'm going to present uh
some slides kind of reviewing
um a recent preprint that is currently
in review
on active inference and the free energy
principle
so i'm going to share my screen
yep looks good okay so yeah i'm going to
give you an introduction to the topic

and then talk a bit about linguistics
and then try and wrap up um
you know we'll make this more of an
obvious conversation uh comments
questions
uh objections obviously uh more than
welcome from anybody
so yeah like i said this is the preprint
that's currently uh in review
uh with emma holmes uh carl first and
ucl uh the argument is that natural
language syntax complies
in some degree with the free energy
principle
so just um outlining some kind of really
cool
uh you know basic principles i want to
make this
conversation a little bit more
philosophical than than normal
and obviously this is this this lab is i
have influence lab so
i want to kind of introduce some
linguistic topics more than the
fap topics and because the audience
might be less familiar with the kind of
synthetic theory so to begin with
natural language syntax
yields what linguists then call an
unbounded array of hierarchically
structured expressions
unbounded meaning that they can
potentially go on forever there's no
upper bound on the limit
of a sentence length right you can
always make any sentence longer by
simply adding
john said that at the beginning of it

and it gets longer the only thing that stops sentences going on forever is the working memory the age of the average human being and indeed the age of the universe you know sentences can't go on forever

they have to stop at some point and but the generative component

in principle could do that so that's the interesting part about language the fact that you could in principle generate an unbounded array of expressions so i kind of argued in this paper that these are used in the surface of active inference

which is in turn in the code with the community principle

so the general goal is to kind of align certain concerns of linguistics with those of the normative model for organic system behavior

associated with the fpp so i'm going to be relying on theoretical linguistics with

special analysis on syntax which is the system that look that determines the kind of order that words

and go in the structure and organization relation the organizational relations between words and sentences

so a lot of these design principles are kind of um

general to the biological component of language not just for specific instantiations of language

so they're not specific to english but french or swahili they're kind of general design principles

um about how the language system seems
to be
like organized and so like i said i want
to kind of emphasis emphasize more
on the linguistics and uh topic more
than
active inference just reverb to the
exposition
and although the pre-print itself has
all the kind of relevant details if
you're interested in reading more
but um yeah but kind of the brief
historical background here is that
since around the 10th century about
20-25 years ago
lingus has basically been and developing
theories of linguistic computation
and which invoke economy principles so
um
basically the idea is that when you pass
an instructor sentence there are certain
computations that enter into that
process
and it's not just a single kind of chunk
that you memorize or produce
you do it in discrete operations
and yeah for some reason there's
currently no means of grounding or
motivating these ideas through more
general non-linguistic domains
so recently proposed principles of
economy um such as
minimal search or least seven criteria
which i'll explain and give you
definitions up soon
and i i argue that they adhere to the
ftp
and if this can be shown then this

permits a greater degree of explanatory power to the fbp with respect to higher language functions

and also presents the language of the first principle grounding of notions pertaining to computability and so on so in other words the idea is this um natural language syntax is a system it's an organic system it's a formal system it can be described

using the the language of uh theory of recursive functions and computability imported from computer science and all the rest of it but there's a problem and all of these principles of economy and language design in the literature are unsurprisingly kind of language focused right they have a very kind of

linguistically encoded background and so there's a bit of paradox because one of the original goals of this program was actually to see

okay let's see how much of language is actually uh

can arise through the kinds of formal principles

that govern the organizing uh shape of you know snowflakes or the mythology of lightning bolts and so on kind of just general

um uh domain general uh laws of nature essentially that can give rise in different ways across different domains and yet the linguist the the teacher is still kind of

uh encoding these in language specific terms

rather than relating them to more
general principles of brain organization
or mental computation
so that's the kind of background and
so many historical insights into syntax
i argue are kind of consistent with the
fbp
which provides a novel perspective under
which the principles governing syntax
are not limited to language
but they actually reflect domain general
processes that underpin a variety of
positive computations
this is also consistent with a strain
within theoretical linguistics that
explores how syntactic computation
may adhere to general principles that
may well fall within extra biological
natural law
in particular considerations of minimal
computation such that certain linguistic
theories might be engaging
with general properties of organic
systems that impact language design so i
think that's kind of a beautiful idea
the idea that
and the language that we speak english
french german
not only are they biologically grounded
but actually the rules that kind of
govern
the uh what you can say what you can't
say and indeed how you how your
brain computes in parts of sentences
part of that process you kind of get for
free if you just assume
um extra biological natural law one of
which you can you can argue is is the

french principle so that's the kind of background here

so um the structuring influence of the fb can be detected in language and not only has been argued recently at the complex level so the um kokkerson and his colleagues have published a bunch of papers in the last year or two mainly in 2020 arguing that and

active inference can relate to narrative comprehension

interpersonal dialogue like when two people talk um cooperative intentional communication

and speech segmentation and but i'm going to kind of argue that all of these things rely on something much more

lower down and that syntax if you can't um

synthetically construct the phrase if you can't put two words together and form a phrase

you're not going to get very far right you at least need to do that in order to engage in

um you know intentional uh cooperation and

narrative and storytelling and speech segmentation

you at least need to be able to form a phrase right if you can't do that you can't do anything to do language so i'm i'm kind of arguing that and

you know all of the all of the ways that the fep can be related to

all of these things here is arises from

all kind of fundamental lower level
consideration
to do with the way that phrasal
computations are executed
um have you changed any slides we can't
see any slides changing
just okay
uh yeah i have changed a few slides just
maybe unshare and reshare
but it stands alone
what you've said as well but it will be
also good to have the visual
you see okay how about this
so we see your mouse and then just
advance like on the left bar
like click on to your next slide yep
perfect
okay what about now no
okay oh i see my bad my bad okay um
well i might have to
like window in focus yeah
this just as a non-linguist was really
interesting that you appealed to like
kind of
bigger laws than linguistics
like complex systems theory or
snowflakes and lightnings like you said
and then yeah just pointing out that
you're kind of
bringing linguistic level rules or
patterns into a bigger
scope okay cool so we see it with the
chomsky now
perfect perfect okay well um i've only
skipped like five slides so everything
i've just said
is so yeah i show the the paper here's
the paper

and i simply bring it to this i exercise

this

all of these things i've already read

out so yeah if you heard what i said

then

that's good great okay so can you see

slide number eight yep

cool awesome okay uh

okay so um if the process of

constructing hierarchically organized

sets of linguistic features into words

and phrases and sensors

can be shown to adhere to principles of

efficient computation then this process

must also operate within saying

fundamental constraints on your dynamics

such as those implied by the fpp to

which the homeostatic brain minimizes

the dispersion

of intraceptic and heteroceptive states

the fpp can also allow us to understand

how natural language complies with the

constraints imposed on worldly

interactions

deriving certain features of language

from first principle so again at the

moment this is all this is a little

pretty abstract

and i'm going to give you some much more

like concrete examples so i apologize if

it's a little bit philosophical

at the moment and so a number of robust

findings from theoretical linguistics

can be used to support the image of the

brain as a constructive organ

assembling and unfixing linguistic

representations in the sense of surprise

minimization and related goals

those syntactic structures are not minor
external entities they're not the kind
of thing that a physicist could easily
examine

but they're rather actively inferred by
the brain uh seemingly with the help of
endogenous slow frequency activity
coordinating cross-court organic
itemization of linguistic features
according to

a bunch of recent models of neural
linguistics which i'm going to get back
to at the end

um so in other words we have a general
framework for how syntax is implemented
at the abstract level

and we also maybe have a kind of a
decent understanding of how that's
implemented in the brain

so that's the kind of the long vision of
this presentation at the end we're gonna
i'm gonna try and wrap it all up by
returning to the brain

um but for now uh i'll give you a more
kind of concrete example

so i said that syntactic structures have
to be unfair they're a form of inference
they're they're inference generation
they're not just um passive

um perception so um if you have a
structure like

we watched a movie with jim carrey that
can mean two things

and it can either mean um the movie
stars jim carrey

the truman show say we watch the truman
show or you actually watch the movie
whilst

um sat next to jim carrey it depends on
how you pass it right it depends on
um how which phrases get merged with um
in in which order basically
and so there's a kind of general roof uh
schematic here
um that you can kind of find a
collaboration of in the pre-print paper
that i mentioned
but the basic idea is that there's
structural ambiguity that arises from
syntax so therefore the whole process of
syntax has to be
an inference process it's not given to
us uh on a spoon we have to kind of
we have to do some homework in order to
construct every possible pass
so that poses the problem how does the
brain do that and it is an inferential
process
so what are the operations involved in
that so here's a
kind of a standard what called free
structures and linguistics
there's there's two different ways of
passing the sentence and you can either
shoot an elephant
who's wearing your pajamas which is
definitely possible or you can shoot an
elephant whilst wearing grandpa jumps
and it depends how you in which order
you merge the phrases right in which
which hierarchical relationship
is exhibited if that makes sense so uh
there's kind of a lot of theoretical
background here in linguistics that i
won't go into because that's a whole
other you know the whole whole different

lecture um but the basic idea is that
there's a
ambiguity in structure generation it's
not always a language is not
um a system of beads on a string it's a
it's a structure
so along along that kind of theme i'm
also assuming
a distinction in linguistics between
what's called i language and e language
and this is a really crucial extension
to get clear so i want to make sure that
um
this makes sense to everybody so an eye
language is
um the actual internal knowledge that an
individual human being has in their mind
brain
i stands to individual internal and
intentional intentional with an s
meaning meaning
that which literally means meaning
generating meanings on the other hand
you have um what i call
the e language perspective and that is
actually
arguably not really a vulnerable or
coherent it's the idea that language is
a kind of external
um a system behind external an extra
mental system
like the english language is somehow out
there in the world and when we learn the
english language we kind of approximate
some kind of mind external system
and we all have something in common i.e
we all approximate to the english
language in different ways

but the island perspective assumes that when we actually communicate with each other and the reason why we can successfully communicate is not because we share an e language in common it's because our languages successfully and sufficiently overlap to the degree that we can actually communicate with so everybody's our language is different it's often said that there's you know seven thousand languages on the planet but that's actually not not true there's actually you know however many people that are on the planet i can't remember it's like seven billion or something less than half and that's how many languages there are and every human being has a different language faculty and it's set that parameters are set differently we all have different um idiotics different understandings and so on and that kind of sounds really obvious right when you think about it it's like obviously that's true but um the implications for the study of language are actually pretty uh kind of natural in other words when we study um linguistic confidence we are studying like a mind internal computational system and we're not studying something outside so the english language is not something that lingua is actually study we don't know that's not a coherent concept

uh it's like the concept of culture or
um community or whatever it's
these are not things that physicists
could you know identify and that's not
to say that we can't say things about
them you know we can't abstract
the human mind count stroke theories of
e-languages i guess you know you can
talk about
the english language you're changing
over the decades
which is you know kind of definitely way
to talk about language and you can study
in that way
but from a naturalistic perspective from
a biological perspective that's not use
the only way we can study it is based on
what an individual mind brain is doing
so to give another example here um if
you take a recent film
that ign released a trailer for
yesterday hitman's wife's bodyguard
and the defining property of language is
said to be this unbounded
array of hierarchical organized
expressions via location
so radication is a defining property of
natural language syntax but it turns out
to be distributed unevenly across the
world's languages
so some people have argued that certain
languages don't exhibit embrocation at
all
and but this actually tells us nothing
about the actual biological language
so in other words speakers of these
languages can easily learn portuguese
right

so the idea is that and language can easily and readily be exhibited by most of the world's languages again when i say words languages i mean a kind of convenient description of every individual's fixed language faculty so most germanic languages only allow a single pronoun and genitive limited to proper nouns but german doesn't allow recursive embedding that possesses though in german you can say john's house beyonce john's sister's friend's house um so the above movie when it's released in germany ashu is going to have to be called something like the bodyguard of hitman's wife which it has a slightly different tone to it so the idea is that um this fundamental capacity of language to execute location and you can find it all over the place you can find it in uh phrase embedding center embedding phrases the way that we actually convert phrases or you can find it in other ways like in this example here so um relating this back to the fbp while the mpp is a variational principle of least action such as those that describe systems we've considered one of these a relatively recent program in linguistics has suggested that natural language syntax adheres to principle principles of least action and minimal search engines

so modern theoretical linguistics
remains comparatively remote from other
fields in cognitive science
but certain postulates from this field
resonate with the fep and it's the
proper
so many biological and cognitive
principles of efficiency might be
special cases of a variational principle
of free energy
pursuing this assessment should allow
researchers from distinct disciplines to
re-evaluate their hypotheses
and empirical evidence in terms of lower
level free energy which is the kind of
goal
i have in mind here but we should also
stress as a recent paper does that the
fap can be used as a methodological
heuristic
for research and but it's not a theory
of everything it's just a framework
and actually i find this very similar to
a framework in linguistics called the
minimalist program
which is a program it's not a coherent
not yet at least
it has a couple of theories working
theories but it's not a coherent body of
doctrine at the moment it's an ongoing
research program with a particular
ideology and framework
and that's exactly what the what the f
is right it's a formal principle
which yields that i can contribute to a
discrete individual and separate
and sort of theories depending on what
domain you're looking at and

so i kind of like the idea they're both
programmatic
notions and and you can implement them
in different ways

so uh so we'll be working here with the
fab

and the paper in question doesn't work
on the fpp so much it's not an applied
approach to the fpp

and so the fab has been argued to be
more of a kind of conceptual
mathematical model for self-organized
systems

and as a recent review by andrews makes
clear there's a number of ways that the
fbp has served as an aid to scientific
work without

constituting like falsifiable statements
about the state of nature

it's a it's program it's it's a way of
usefully describing

uh certain uh joints of nature without
necessarily carving them

so while we argue that natural language
syntax and so when we argue that natural
language syntax complies with the free
entry principle

i'm not necessarily implying that the
fab you know necessarily bears

specific direct predictions from the
behavior it's rather way of motivating
the construction of neural

novel conceptual arguments for how some
property of organic systems

might be seen as realizing with those
so as i said uh repeating their language
language distinction and focusing on
knowledge which is internal to the mind

of the speaker
exploring their apparent competence
rather than you know what they happen to
produce
so one example of what you can think of
immediately is the existence of
syntactic structures
can contribute to a unique form of
episode encouraging through maximizing
model evidence and minimizing surprise
and variation of energy the actual paper
in question that i'm i'm citing here
kind of goes into more detail but the
basic idea is that
the highly restricted uh set of
syntactic projections while called
um the way that you can categorize a
phrase and nouns and verbs and
complement as a brace and so on that
restricted set
that finite set of ways that you can
define a particular phrase as being a
bit of phrasal vampires
and achieves that goal so you can also
assume that languages
and language users during comprehension
select the phrase structure that's least
surprising from the perspective of the
hierarchical gender model
again this goes back to the um we
watched a movie with jim carrey example
it's a prediction and uh uh plays a
pretty big role here
so our search for a structure building
can be cast as an internal action it's
an actual internal
action that an individual does in the
sense of the active inference framework

that selects someone competing
hypotheses i.e syntactic
structures you can pass it this way you
can pass it another way
and also the same goes for lexical
semantics not just syntax
and in other words you can you can you
know interpret an individual way of one
by one
and i'm going to give some examples
later so pretty much all of language
from individual words up to sentences
and are inference generated
and also a little side point here and
relating more directly to existing work
on active inference and communication
and it's i think it's pretty interesting
to know that the recursive combinatorial
apparatus of
syntax has been argued to facilitate
recursive theory of mind right
so the ability your ability to know that
someone else knows that she knows like
you know this and so on
and it could be seen there for his
deriving or piggybacking in some way
and active influence-based properties of
higher order linguistic communication
and which in turn serves to unveil the
latent or hidden states that are other
people's mental states
which is the kind of standard assumption
and theory of mind active inference
research but i think it can be related
much more directly to language once we
understand
that the extended kind of uh uh
use of our theory of mind in language is

it comes directly from this because
recursive property right
okay so i mean i'm going to move on to
kind of the more concrete examples
everything i've said so far is pretty
philosophical and pretty
abstract but i want to kind of give you
some actual concrete examples here so it
makes a bit more sense
so free energy provides additional
constraints on what a computational
system can be physically realized as
which is very useful
so take the first three principles in
classical recursive function theory
which allow functions to compose
substitution primitive recursion and
minimization these are all designed in a
way that you might think of as
computationally efficient
they reuse the output of earlier
computations uh so for instance
substitution replaces the argument of
function with a function primitive
recursion defines a new function on the
basis of repairs of call to itself
bottoming out and a previously defined
function and minimization produces the
output of a function
with the smallest number of possible
steps
so the notion of minimizing surprise can
be used to ground observations from
theoretical linguistics pertaining to
the grammar's propensity to
reduce the search space during syntactic
derivations permit no tampering of
objects during spectacular derivations

so it restricts the number of resources
able to be combined in a given moment so
in other words this is very
kind of an abstract example but i'm
going to give you some more concrete
ones soon

uh during any particular part of a sense
you don't just search the lexicon and
extract a new unit just because you
can't you only do so if you need to
if the current and number of lexical
items that you've already said so that's
gone for
and suffice to generate a given
interpretation then you don't need to
search for more items that's sufficient
right so in other words don't just do
don't do more search when less search is
possible

and also relates to the grammar
principle to limit the range of
representation of resources
able to be called upon during any given
stage of comprehension

so examining some core principles of
occasional natural language
clearly exhibits minimization right
while binary branching of structures
limits redundant computation so binary
branching

uh calls back to that slide i showed a
few minutes ago where you have these uh
binary branching pre-structures so i
shot an elephant in my pajamas
and that all uh relates to that can will
be derived through successive
implementations of a binary branching
operation that simply puts two things

together and gives it a second identity
so natural language syntax also exhibits
discrete units obviously there's
individual words and which leads to a
discreteness continuity
so syntax is driven by closeness of
computation so objects x and y
form a distinct object x and y its
objects are also bounded so there's
a fixed list so there's nouns verbs
adjectives adverbs prepositions
converses
uh tense objects uh little nouns and
little verbs
aspectual elements and so on all these
different particular functional
and content related notions their
hierarchical ordering is based on a
specific
functional sequence as well and which
imposes direct restrictions of
combinatorics
these objects can be combined into
cycles so recursive embedding
which can be extended to form non-local
dependencies so an example of that is
you have like number um plurality
marketing
so you can say the keys to the old
wooden cabinet are on the table
um not the keys to the old wooden
cabinet is on the table because
obviously it is those are keys
that they have to mark but they have to
agree in the number feature right
so that's an example where you have a
non-local and dependency between two
elements

so you have to hold it in memory uh even
though they cross
phrase boundaries so syntax still keeps
on generating new structures
but you have to associate one element
with a new element further down the road
and so these properties are in turn
guided by principles of minimal search
unleashed effort
which i'll show in a minute fulfilling
the goal of active imprints to destroy
meaningful representations as efficient
as possible and i think that's really
what
it comes down to that's the kind of core
crucial message here
and if it can be shown that the the
language system
uh constructs meaningful representations
as efficiently as possible then
therefore it must be on accord
with the fdp and again that contributes
to surprise minimization amongst other
goals
so from the perspective of fep and the
range of possible
structures available to comprehenders
provide alternative hypotheses
that generalize as such preclude
overfitting sensory data
so if the complexities of linguistic
stimuli can be efficiently mapped to a
small series of regular
and regularized in the sense of blend
syntactic formats
this contributes to the brain's more
general goal of restricting itself to a
limited number of characteristic states

so in other words only change you
believe about things if you have to
right
and by mapping syntactic structures to
language external conceptual interfaces
now that's a kind of a key thing
language external
interfaces what that means is you have a
narrow
component of language the kind of narrow
what's called the narrow faculty of
language we can call it whatever you
like it's just the capacity to construct
phrase structures
but then you have the language external
mental modules
and memory systems and what have you and
attentional control systems uh in the
mind that are located across different
political
surfaces you know it's it's the kind of
standard the standard modular framework
i guess
and so by mapping these structures that
i've been generating
language to these language external
interfaces
in a minor adhere into principles of
economy language can be seen as engaging
in a series of questions and answers
with sensory data
specific so all the recent work in
theoretical semantics assumes that there
are
and there are fetchable concepts that
language can interface with
and they're also non-fetchable concepts
so in other words that there are systems

that
there are properties of human thought
that the language
system seems pretty keen on um and
it likes to use and exploit but there
are other uh
modules of cognition that for some
reason um can't be linguistically
encoded as easily
that's kind of a weird property right
like why should we why should we be able
to linguistically encode
and communicate about certain thoughts
from others
and so one example is many of the
world's languages again
when i say words languages i mean that
in quotation marks
and what i really mean is many of the
individual human beings
they make use of quantification or
numerosity often you know
typically seen as via this frontal
parietal
uh quantification network which is
interestingly closely linked
uh to major language sites so that might
be one of the reasons why
it makes use of it and but they make
little use of color
despite color featuring just as
prominently an ordinary experience right
our sensorian is filled with colours
um and this might be due to the
remoteness of occipital visual regions
in the brain
maybe maybe not uh that's one
explanation maybe it's wrong

but it's one possible reason so for instance one might imagine some functional morphine um encoding brightness or shade of coloration in language you can imagine it like if you were to invent a new language you could you could encode color uh features somehow or some kind of inflectional but that doesn't seem to be the case even though like i said that color is a pretty important part of your life at the same time language seems to make considerable use of certain contentful concepts like eventuality but not others like worry if you're like you know no things mark how concerned somebody is about something i'm very concerned i'm a little bit concerned i'm not very concerned which seems to be making potentially neurally efficient use of specific access easily accessible representational resources rather than less easily accessible cognitive modules right to map complex meaning onto natural language expressions so it's fairly observed that language acts as an artificial context which helps constrain what representations are recruited and what impact they have on reasoning and inference so words themselves are

highly flexible
and metabolically cheap sources of price
throughout the neural hierarchy this is
a really cool idea that i'm going to
expand on
in a few moments of time so to give you
some more examples of
again writing this back to active
imprints and take
the second ball second blue ball to the
left of the large box
that's a pretty simple spatial direction
but it can only be encoded
via natural language syntax right you
need that kind of
hierarchically organized phrase
structure to generate that particular
thought it's a very simple thought
very simple direction but you can only
generate inferences about it through
using the recursive
um combinatorial apparatus of language
similarly you have structures like
the young happy eager either going to
oxford cambridge pleasant man
this involves unbounded unstructured
coordination involving
disjunction too so the disjunction is
you know x or y
uh which is a highly complex structure
to compute it's a highly complex
conceptual structure and your language
users can easily do it and readily infer
it but it also um opens the
the door for a whole new species of
inferences to be generated
new thoughts about the world new
possible hidden states new things that

you can think about
and one of the examples of jerry photo
or the philosophies together
and uh he said you know if you take a
a a pen and a piece of paper you can
easily draw
a man you can easily draw a zebra but
it's kind of
difficult to draw the thought that there
is not a zebra next to you
right so if you try and conceptualize
the thought there is not a zebra next to
me
how can you draw that how can you depict
the fact that there is no zebra next to
me
that's kind of weird it's kind of
difficult to do i suppose you could draw
like
a man and then a zebra and a line
through the zebra well
that's kind of weird that still
presupposes that there's a zebra next to
him that you reject
um so language uh generates a whole new
species of
a new format for thought basically it
generates a new format for thought
and which seems to be unique in other
words not readily
and translatable into other domains like
visual
representations or so these rapid
inferences are our properties and states
hidden states can be generated
relatively effortlessly by language
and like i said no other computational
system and human cognition

can achieve with this well that's the idea

so a number of economy principles have been proposed in theoretical linguistics these are all very you know kind of technical um syntax related notions so i'm not going to explain too many of them but i will give some examples and these have all been framed like i said at the beginning they've all been framed exclusively within a linguistic context

invoking highly domain specific notions despite a core part of the intended project

of modern theoretical linguistics being to embed linguistic theory within principles general to cognition right so for example the inclusiveness condition maintains that no new elements can be introduced

in the course of a particular syntactic derivation so once you integrate once you're passing a particular sentence and you're deriving

a particular representation of that you're not just going to randomly introduce a new lexical item or a new um hidden element just for the sake of it you're only going to do it if it contributes to immediate interpretation and so only existing elements can be rearranged restricting available resources

but it's also unclear to what extent this computational principle finds analogous examples in non-linguistic domains right

so one way of motivating these
language-specific and generalizations
by making direct reference to the fpp
will
not only foster i think uh more crucial
relationships between
their theories of hypocrision and
neurobiology but will also broaden the
explanatory scope for the existence and
prevalence of particular syntactic
phenomena
and but what's interesting to know is
that linguists readily admit to lacking
a specific theory of computational
appreciative language
like i said at the beginning it's kind
of a program right it's a programmatic
notion
so in a recent paper galeo and chomsky
point out to be sure we do not have a
general theory of computational
efficiency
but we do have some observations that
are pretty obvious and should be part of
that period
right that's basically the state of the
field uh linguists have very
um uh well uh theories of uh
language specific uh efficient uh
efficiency
uh criteria but they're translating that
into kind of more domain general
uh gold areas has not been so successful
but we can at least suppose that
whatever definition will be forthcoming
will be related to more generic notions
of economy like hamiltonian notions
or minimizing energy expenditure during

language processing uh shortening the
description of your minimal description
uh reducing pomegranate complexity and
the degree of necessitated belief
updating again like i said you know
requires one's belief
if needed and more happy
so one of these uh so-called
minimal computational procedures is uh
what's called relative rise minimality
so this is the principle that states
that given a particular configuration
x z and y a local relation cannot
connect
x and y if z intervenes and z fully
matches the specification of x and y
in terms of the regular features i'll
give an example of that so in other
words if x and y attempt to establish a
syntactic relation if you go if you call
back to the long distance dependency
thing i mentioned
and put some element into beans and
provide and can also provide a super set
of extra particular features ie x's
features plus some other features
this box of relation that sounds
extremely abstract
so i do apologize for that but this is a
more concrete example
so in the sentence you haven't won and
which game provides a subset
of the features um hosted by howe
which results in unacceptability but the
equivalent does not obtain in two
and so a relationship between both
copies of which game can obtain
licensing interpretation uh and the

strikethrough

denotes the originally made position
after uh after movement has taken place
so it's originally merged down here
and then you move the structure to the
beginning of the sentence in order
to form a question so question formation
often uh involves just moving all in
uh in the middle of the end of the
sentence the beginning to form a
question

and so if you say how do you wonder
which game to play
that sounds pretty ungrammatical but if
you say which game
do you wonder how to play which has the
same approximate interpretation that's
okay

and the reason why is because uh uh
howe hosts only a queue feature a
question feature
and as it searches down the structure um
it

encounters which game now which game
bears a cue feature because of which
but it also has a now feature because of
game and so therefore it can't reach its
final destination

the final destination being it's
originally made position after play
which only has a queue feature
on the other hand if you move uh which
game to work do you wonder
and you leave howe in situ then which
game carries these two features here
and it skips over how because how does
does not satisfy
the full feature specifications of which

game right so it needs to search further
and then only when it searches back and
reaches to its originally merged
position
does it interpret it so this mean what
this means is when you say
which game do you wonder how to play
you're interpreting which game at the
position of play
because you're not into you're not
definitely that edition of wonder you're
not asking which game
do you wonder how you wonder you're
asking which game do you wonder how you
play
it's all about the question is about the
plays on it
and so this has been argued to emerge
directly from minimal search
uh allowing this higher level
representational principle to make
directly from properties of official
computation this is one example where
you have a kind of a
rhetorically kind of gross higher order
uh principle in linguistics being
reduced to a lower level and more simple
kind of element
so translating that into minimal search
you cannot even consider
as i just said when you search the
structure for matching your features in
two
the minimal stage procedure would simply
skip out but it will find the original
copy of which game
because it's searching in across the
full structure

so another example of economy can be found in the principle of full interpretation which simply states and that there are no superior symbols allowed at the two linguistic interfaces so these interfaces are assumed to be the conceptual and sensory window systems so in other words the two things you can do with a linguistic structure is you can interpret it or you can externalize it you can produce it you can say it you can sign it uh you can translate it into braille and so on and or you can simply think it so this ensures that the system need not compute symbols that are ultimately subverted to the goals of either interpretation or externalization so for instance in three this has an argument that does not have a semantic role so therefore it's unacceptable so you can imagine um the sentence walt gave jesse a gun that's fine all the semantic roles are filled there's an agent there's a patient and there's an instrument being involved the particular tool but then if you add to saw there's an additional kind of location preposition and being marked there but it doesn't have a semantic role so it can't be interpreted and that sounds kind of trivial right like most people when you when you give them these examples they

say well yeah obviously that's kind of
not a very deep thing
but actually it's it's a pretty puzzling
like phenomenon it demands actually it
has to be explained somehow
and so one of the operations that's been
involved in this uh
minimal computational procedure is
called merge
so the operation merge simply takes x
and y and forms the set $x \cup y$
it just puts two things together and
this constructs the binary branching
structures that i mentioned
but merge itself can also derive some
core uh separated properties of
linguistic relations
such as membership uh dominate team off
and so on these different hierarchical
relations between nodes
and in a tree uh you know one one uh
branch of a tree being higher
or more deeply embedded than another one
and as well as
other relations see what's called c
command we can put that aside
and so in brief much of the complexities
of syntactic relations and can be
derived from successive instances of
this simple merge computation
and reducing complex visibles to simple
invisibles
so for instance if you the example i
gave here with which game
you can imagine uh the set $a \cup b$ being
constructed
and then if you take the same element b
and you merge it with the set again

you get b a b and then you can delete
the first appearance of b
and then you simply get the linear order
would be b a
right rather than the actual set being
interpreted as b
a b and this seems to be what's
happening here where you have a b
over here which game and then an a the
whole structure
and then b again being originally
measured
so all of these kind of con
superficially complex
linguistic phenomena which on the
surface seem very complicated and very
elaborate
they can all be boiled down to a very
simple operation which is just
take two things and put them together uh
and
what's interesting about this is that
related when i'm not like a junction uh
do not involve modification of the
semantic content of the structure the
adjunct is a candidate room so an
adjunct is something like uh
like a prepositional object like in the
path or to the beach
and so for instance if you think a
structure like john and mary talked in
the bar
the fact that they talked in the past
doesn't change the fact that they talk
right so the actual original
interpretation the syntactic structure
of the integrity and the meaning of john
and mary talked

is not changed by adding an adjunct so
you can add an adjunct in the park
but that doesn't change the fact that
john and mary still talk right
again that sounds like a very trivial
property it's obvious
right but again it demands explanation
it has to be a reason for that and the
reason why is it turns out that and
when you add an adjunct you simply um
you simply linearly add it to the end
you don't change the actual identity of
the phrase itself
uh there's no what's called labeling
there's no labeling involved
in prepositional major merging so you
don't change the identity of the pocket
rates
so the ftp has been equated with the
principle of least effort
and its process theory is active
inference so strictly speaking the fpp
basically is a computational principle
right
uh the probabilistic beliefs it's
concerned with are directed absolutely
namely external states of a
self-organizing system and in in a
similar way that the entity is a
researcher is that
so too is much of recent theoretical
linguistics guided by programmatic
concerns right like i said it's a
program it's an ongoing kind of ideology
on the other hand linguists have
developed theories of syntactic least
effort like i said
um but the process theory is a little

bit less clear right how it's actually
implemented

is slightly less clear and what i would
argue may become more clear

if it can be accommodated within
existing frameworks and knowledge
that the active infrastructure can bring
with it to help

solve the puzzle of how a language is uh
implemented in the brain

so here's another example for you and
routinely poems that rhyme evaporate

so in this instance routinely
exclusively modifiers evaporate

so the wave routinely goes with
evaporate that's how they're injected

so it can't modify rhyme even though
this word is closer in terms of linear
distance to routinely right

so ryan comes one two three words after
but evaporate comes one

two three four in terms of linear
distance so they're more

they're more linearly remote and yet in
terms of interpretation

they go together and the reason is and
the matrix predicate

evaporate is closer in terms of
structural distance to routinely
and the reason why is um that rhyme is
embedded within

the phrase headed by poems so it has
it's on it exists on a different

hierarchical plane if you like it's kind
of lower down in the hierarchy

and evaporate and so language computes
over structural distance and not linear
distance

so sensors are not simply beads on a string they're not linear objects and they have to be linear objects in terms of centromere externalization because we live in the universe we live in we can't we can't speak in parallel we can only speak in linear linearization although there is some evidence that uh sign languages uh can communicate somewhat to some degree in parallel like you sign one uh element with one hand and then another word so there's some evidence that sign language might be able to defy the laws of physics but it turns out it's that's probably exaggerated to an extent it is still a form of linearization but just kind of a co-linearization development so language prioritizes the demands of the syntax semantics interface over other subsystems like morphophonology so while two structures might exhibit different linear orders they may exhibit the same underlying hierarchy so here's a really good example in english and fast the verb direct object dependencies are the opposite but the interpretation is concerned so john has read the book you have john then auxiliary then the then the object and in basque you have a different order you have john then the book then the

verb and then the auxiliary
and yet they mean the same thing they
have the same interpretation right
so this suggests that um a kind of more
fundamental
uh operations going on here namely
syntax encodes the verb and direct
object
as an abstract phrase which omits the
subject so in other words in the syntax
in english and basque you have the same
underlying syntax which is
subject and then verb and direct object
being merged so you merge the b and do
first
and then you accommodate the subject and
then when you linearize that
when you you know communicate it in the
externally and you do it in different
ways in english you do it one way
in past you do it another way and but
the basic idea is the same you have the
same
underlying interpretation that also
accounts for something pretty obvious
mainly the fact that you can translate
you know one sentence into another
language right
that's that's a fairly obvious thing
that you can do with language and so
therefore there has to be some kind of
commonality somehow
but the commonalities might be much more
deeper down than most people kind of
appreciate
so through the various stages of
language development as well innocent
children

don't typically produce expressions that deviate from general grammatical principles pertaining to the structure dependence of rules even when they produce so-called mistakes so there's been a lot of research on a child language development so in other words when children do make mistakes they seem to make mistakes which accord with the grammatical rules of their language which suggests that and sensitivity to um structure dependence uh forms a core part of language design so corpus studies of infinite language exposure reveal that there are actually very few digrams let alone trigrams so statistical procedures can help but there seems to be some more kind of innate sensitivity to structure dependence which seems necessary and as a recent paper also reviews human learners prepare to induce hypotheses that have a shorter description length and logic with simplicity preferences possibly being a governing principle uh of uh cognitive systems all in the code of what the fpp would predict so simplicity-based preferences i've got a range of formal language models too relating to the notion of minimal description and you know you might also invoke principles of minimum redundancy so this is kind of a really important idea

uh minimal computational efficiency
seems to be a really general
cognitive goal of the brain and there's
a couple of recent papers one of them in
text i think that came out
maybe last year or maybe this year i
can't remember um i think was called
memory as a computational resource
which showed that um across a bunch of
domains human memory in its various
guises
also exhibits uh an adherence to
principles of efficiency
so i think it's not too surprising to
when linguists come along and say
you know language also adheres to
principles of efficient computation
so all of these ideas seem to be you
know out there at the moment
and everyone's kind of coming to more or
less the same conclusions but
just using different language just using
different background assumptions
but the general idea i think is is kind
of
they all mesh well together so as i said
linguistic computation seems to be
optimized for the generation of
interpretable structures
rather than for the generation of
maximally communicative messages
on specifics so in other words and
whenever there's a conflict between
principles of computational efficiency
on the one hand
and principles of communicative clarity
on the other performer typically wins
now this is not to say that when we do

communicate with each other as we're by
first and then
when we do communicate we do it in an
efficient way but that's a separate
question from where
the language system is designed in a way
as to maximize
that communication so the normal
functioning syntax seems to lead the
instances which reduce
communicative efficiency and prioritize
inference generation
so that so the goal of the language
system is to generate
particular inferences and
representations about the environment in
an efficient way so here's a pretty
clear example of this right
if you take the sentence you persuaded
saul to sell his car
the individual and the object can both
be questioned but questioning the more
deeply embedded object
in terms of the hierarchical structure
forces the speaker to produce a more
complex safe locution
right so you can say uh who did you
persuade to sell what
but you can't say what did you persuade
who to sell even though they mean the
same thing
right same word same interpretation all
it means is
you know what is the individual and
what's the object that's it
just just tell me who the individual is
and what your objectives
but you can only say it if you construct

it in
the most computationally efficient way i
search for the first possible element to
question
right if you search for the more deeply
embedded object
can't do it and so the structures in 11
involve the same words same
interpretations
yeah the more computationally costly
process can't be licensed so this is a
pretty good example and there's plenty
of examples like this by the way i've
written a paper about it in gloucester
plenty of examples in which there is a
clear conflict between um
syntactic uh priorities of just
generating a meaningful structure
and generating possible structures that
would actually aid
communicative efficiency and
communicative flexibility that's not a
priority of language
so other examples show that the
acceptability of sentences can be
impacted based on the extent
to which the construction makes and it
contributes to a novel
non-redundant contribution to one's
mental models beliefs again this is
really
directly in the core of what active
entrance would predict uh rather than
those of course specifics
again reinforcing the role of syntactic
processing and inference generation
rather than communication
so the greater acceptability in 12b that

the real by the way the reason why these
are number 12 13 is that's that's that's
the numeration in the in the paper
so the reason why 12b uh seems degraded
relative to 12a

seems to then from the fact that the
speakers are unlikely to be ignorant of
the relevant content right it's kind of
a pragmatic reason

so kim knows where the soul's in bed
that sounds okay but kim knows whether
i'm in bed

sounds kind of weird even though it's a
technically grammatical sentence

uh it sounds weird because you would
never say it it doesn't contribute
meaningfully to

revising or contributing to once mental
models or beliefs about the world
so therefore the language system doesn't
like it which again

brings in closer contact uh language
design with the fdp

and also cases such as 13 real how even
processes such as contraction
are sensitive to hierarchical structure
and can't be executed over any other
and any random word boundary so you can
say source taller than chemist
but you can't say source taller than
kins and the reason is because there's a
invisible phrase boundary between those
two elements

other examples that related to it are in
14 and 15 you can say

what do you want to do and you can
contract and say what do you want to do
but if you say who do you want to read

the book and you can't contract that to
generate

dude you want to read the book that
sounds a bit weird i mean you
technically can't say it
right and if you say that to me i would
know what you mean i know
you mean straight away there's no
problem but it sounds a bit more awkward
again the idea is that there's an
invisible phrase behind you there but
commits that stops contraction
uh efficient computation uh or at least
structured dependence actually i should
say

is also exhibited in more classical
examples in the literature
so if you say the man is happy you can
uh question that
that structure by moving the auxiliary
to the front
and saying is the man happy and so you
might be forced to conclude well maybe
to form a question you simply search the
structure and take the first possible
element but that turns out not to be
sufficient

so you can say the man who is tall is
happy but you can't say is the man who
tall is happy
uh because who is tall is again similar
to poems like ryan
routinely is embedded more deeply in the
the man-headed phrase
is that the the question uh is happy
which is higher of the hierarchy and
easier to search for
so therefore you have to say is the man

who is tall happy because it is
is actually closer to the element you're
questioning in terms of structural
distance
than the ears in who is tall so again
syntax cares about structural proximity
and i'm not linear proximity
and there are also constraints on this
as well and so you can say john a
chicken and bread for lunch
um and you can question the whole uh
phrasal conjunct chicken and bread you
can say what did johnny punch
but if you want to efficiently question
one element
that's how you already knew that giant
chicken but you're not sure what else he
ate
you can't say what did johnny chicken
and lunch which
there's no reason why you can't do that
right like why can't you say that it's a
terribly fine thought
and you already know that i ate chicken
you know someone just told you ate
chicken and something else
um but you can't say what the giant
chicken had for lunch
because the whole the syntax respects
the integrity of the phrase chicken and
bread
it's it's it has a phrasal identity that
syntax respects
and you can't violate it you can't just
violate the phrase boundary and only
interrogate while
other examples relate across phrase
boundaries and confidence boundaries not

just within them

so you can say sam gave a guitar to me

and loaned a trumpet to you

and you can question both elements you

can say what did sam give to me and loan

to you

but you can't say what did sound give to

me and loan the trumpet to you

even though again in terms of

communicative efficiency that's a pretty

simple structure to generate you already

know that he loaned the trumpet to you

you want to figure out what sam gave to

me right

and again these uh relations of

hierarchy

you can find them all over the place uh

so pronoun uh

reference is a pretty good example you

can say mary said that he has a lot of

talent

and that peter should go far in which

case the pronoun he is being connected

with peter

uh in which case you have the pronoun

here

coming before the actual element peter

but then when you simply take that

phrase and question it and unstate it

it's no longer acceptable

you can't say he has a lot of talent and

peter should go far that sounds a bit

strange

he should refer to someone like john and

and the reason why is because when you

embed

that structure in one a large structure

it changes the actual identity of the

conjunct

so the conjunct that headed by that is a
compromised phrase

whereas the conjunct headed by he and
peter are simply tense phrases

and other puzzles exist here as well so
you can say john

you can say john said he is proud of his
house in which case

he goes with john but it sounds weird to
say in john's house he organized a
meeting

and when he refers to john again you can
kind of pass it you can force it but it
sounds a bit more awkward

it's more natural for he to refer to
peter if you say in john's house you
know someone else

that organized anything and the reason
why is because code reference
via this phenomenal phrase blunting is
bad since syntax preserves

interpretation across movement so the
original structure that's generated
is uh john said is he's in john's house
he organized a meeting right um that's
that's

generated from a more original structure
uh he organized the meeting of john's
house

in which case you have uh he uh coming
before john in the same kind of
tense-free structure that i mentioned
earlier

so in other words syntax seems to win
over linear precedence

although kind of quick examples like
this too so you can say i gave her the

book that sarah always wanted
but if you say i gave her the book that
sarah wanted but again that sounds
slightly strange
changing the syntax by adding the
adverbial element changes the actual
uh content of the phrase itself which
allows more easier co-reference
so um stepping back a little bit this
this whole framework of like merging and
generating hierarchical structures
has been argued in the literature to
kind of uh boil down
from a more domain general lower level
uh computational procedure so some
people have called it the universal
generative faculty
which is just the ability to construct
hierarchical structures and map onto
different interfaces
so the idea is that and when we have our
system of moral
judgment formation which involves agents
patients events and so on
that still requires some kind of
combinatorial apparatus to generate
those judgments
same with music it's been known since
the 70s that um
musical structures have a kind of higher
up correlation to them
and then also with a numerosity with
numbers you can imagine that it's been
hypothesized by chomsky
that if you restrict this operational
mage to a single element and simply
reapply it you can kind of generate the
natural numbers right so you can kind of

form the empty set and then merge it with itself and then merge it with that object and so on and that kind of gives it uh you can call it zero and then call the next one one and call the one after that too that gives you the natural numbers but the general idea is that you have an underlying generative and faculty that can interface with different subsystems so in interfaces with the sound system you get music when it interfaces with whatever morality is theory of minds structures or judgment formation or whatever you get moral judgments uh when it interfaces with the system of quantification and numerosity you get the natural numbers and then when it interfaces with the lexicon whatever that is even more mysteriously you get language which is exactly what the album i'm saying right here so huge f major sound music and so on and but interestingly only language seems to attribute to these most elements an independent identity so with quantification music and morality you simply involve a generation of a chunk some kind of chunking that's happening but with a natural language you seem to get an additional operation you don't just chunk things you chunk them and then you give it an additional identity right you give it

a kind of the the um the sum is greater
than the parts i guess you can call it
that that way
so and what's interesting is that you
can use that merge structure to then
call it again
so kind of you can kind of treat it as
an independent object
independent of its constituent parts
whereas you can't you don't really seem
to be able to do that in music or
language
uh sorry uh non-music domains so there's
also um
a recent paper uh by stan duhain who
argues that this
capacity to generate nested tree
structures um is a human specific
kind of species property right and it
gives a bunch of different um
neurobiological instantiations with this
and but the basic idea is that you know
only humans have language
uh along alongside that you get an
interest in
a unique level of tool complexity
possibly due to this linguistic capacity
so for instance um spheres are
constructed from a rock and a shaft
but it's not just a sphere it's not just
a rock plus a shaft right a sphere is a
rock muscle shaft
with an additional abstraction the
functional use of it the utility
i'm going to come back to that in a
couple of minutes but uh language seems
to be uniquely
concerned with functional abstractions

like use not just form
and then on top of that it seems that
only humans have what chomsky has
loosely called
the science bombing faculty so only
humans are scientists
not surprisingly um we can do things
like peace and abduction or
non-deductive complex infant generation
where you get you know some weird event
ear cares but then we posit that if
a is true and then yeah e would simply
follow naturally
right so we assume a and then of course
on top of that we have the other example
i mentioned theory of mind boxes by a
hierarchical language you know i know
that you know that tune
so there's all these kinds of weird
human specific cognitive traits
all of which can potentially maybe be
boiled down to or connected to
some linguistic capacity so here's a
good example that the hane gives
and these are five different types of
sequence sequences that you can generate
only human beings seen capable of
generating nested tree structures
as i've argued here and in the paper and
we generate tree structures not just
randomly
we don't just do it because we feel like
it we do it efficiently and in the
for the explicit purpose of active input
generation
so transition and timing chunking
ordinal knowledge
algebraic patterns these are all

non-human uh capacities too
bonobos macaques fade song they all
exhibit these forms of things
it's only humans that can do uh option
number five the nested true structure
business
so again this is this is the idea that
when you merge car factory
you don't just manage two nouns right
car and factory
you actually create a noun phrase a
structure that's bigger than the two
parts
so in other words the factory is not in
our phrase and car is not a nanophrase
you need both of them to create a noun
phrase and then you can use that noun
phrase as an independent unit
um with a certain kind of computational
identity so this leads to a kind of a
more
um important question i think which is
what is language
all human beings have language we all
have very strong opinions about what
languages
and but consider the fact that geometry
was originally the study of land
measurement
right back in the back in the day but
developed a sufficiently rich body of
knowledge to abstract away from original
its original object of inquiry and
departed also from common sense
intuition
so our common sense intuitions about
what languages actually have no place in
science

uh data for common sense notions of you
know massive energy of physics
so mit and professor ed federenko
recently conducted a mechanical tech
survey a study asking ordinary people
what they thought language's primary
function was
uh now most of them said communication
uh in line with common sense
and she used this um this data to
criticize
the uh idea from a saying part of
linguistics that language isn't
basically an instrument before
its primary purpose is to contribute to
conceptualization
and but a physicist obviously wouldn't
conduct a mechanical
survey randomly asking people what they
thought the nature of light is
and a biologist wouldn't concern
themselves with people's intuitions
about how hearts and livers were
um and so natural language syntax i
think should be investigated using the
same
standards of scientific inquiry as any
other object in the organic world right
there's no reason why people's
intuitions about language
um should we need it right in fact if
that were the case would just there's no
need for linguistic departments let's
just
ask a bunch of random people on the
street uh that's that's all we need to
do
so on top of syntactic phrase generation

and we can also frame this as
contributing to policies
uh used to perform particular free and
free energy minimizing actions and not
just generating you know
uh linguistic objects so the rapid and
reflexive identification of objects
states and the refs in the in the
external world through simple linguistic
means
uh can yield complex flexible
interpretations for some of the most
common nominal nominals is just a fancy
word for
nouns aiding in the successful
generation of internal models of the
environment
using a limited number of resources and
so objects in the world have to be
identified and they have to be
identified now
immediately right in order to be
successful in in navigating the world
you have to understand things straight
away
and rapidly identify things that's kind
of obvious
and but at least the certain puzzles are
so for instance complex forms of what's
called palisami personally just means a
weird having
multiple senses and and politically also
turns out to be much more widespread
than most people think
it turns out i am almost i remember i
think half the way is in the oadlessness
and complex forms of i think was 46
percent

uh complex forms of policy generated
via multi-wave constructions
allowed for a more precise and exact
localization in conceptual space
than discrete symbols science and
gestures right with natural language
syntax allowing the generation
of a more accurate unveiling of hidden
states in the world
so natural language syntax allows us to
more accurately position ourselves
in conceptual estate space right again i
gave the example of the second blue box
to the left well here's another example
for you
you can say the poorly written newspaper
that i held this morning
has been sued by the government that's a
perfectly fine sense
but it's referring to three different
senses of a simple word like newspaper
so a newspaper can simultaneously be a
piece of information
it can be a physical object it can be an
abstract organization
and we can also call upon all of these
senses at once and yet notice that
this sentence cannot possibly refer to
anything in the world right there's
nothing this is not a kind of thing that
a physicist could explore
and something that's poorly written
something that you hold something that
can be sued it's not a coherent entity
and yet language allows this um simple
policymaker's word
one single lexical item to generate a
very rich

range of perspectives to interpret
experience which is exactly what you
would expect from the active
infrastructure
and so since there can't possibly be any
object in the external world that
complex business word like newspaper can
index

a one-to-one mapping with another
framework we are developing here
lexical items could partially be seen as
hypotheses about the structure of likely
co-occurring sensory input right or
hypotheses about
ontological and meteorological relations
between objects and states in the world
so in other words a word is not simply
something that has a conceptual meaning
a word does not simply fetch concept and
a word is basically an upper eye
hypothesis

about what the world is what we can
interpret experience to be
in any given moment and we basically
test the hypothesis
so we use the word newspaper as a
hypothesis about what's going on outside
and maybe it fails maybe it succeeds it
depends on the context it depends on our
state of mind

and also depends on our interests and
our concerns

i'll give some more examples uh in a
second just to kind of illustrate that
so a recent paper by paul christian's
lab headed by the amica setter
2020 in the paper in frontiers they note
that from the perspective

of active inference things only exist as
a label
or hypothesis or inference about hidden
states
so the contention that i'm kind of
presenting here is that forms of complex
meaning
derive from natural language semantics
form a core component of this
labeling mechanism in active interest so
linguists like to talk about you know
lexical items
both book table walk and so on these are
basically just hypotheses composed from
distinct
core knowledge systems in the mind so
our sense of geometry our sense of place
our sense of social relations
which can elucidate environmental
regularities essential to active
inference
so here's some more examples and again a
nice little quote from
that they make us the labeling quote and
let's take the
notion of vagueness so i was once an
infant
um but i'm no longer an infant but i'm
still me
right and the boundary between infancy
and childhood and childhood and adults
there are legal terms for that but
that's kind of a just arbitrary choice
the actual concept of infancy is an
arbitrary boundary
uh um some philosophers do actually
think that there is a specific
nanosecond

which transitions you from infancy to
childhood but
you know i i think that's unlikely i
think applying
fixed quantified uh uh notions like that
to uh
intentionally an inherently vague notion
like infancy is kind of a paradox it's
meaningless
you know infancy is just infancy it's
not meant to be a precise boundary
uh unless you're a legal scholar in
which case that's fine you can define
legal
legal boundaries between things but
that's kind of irrelevant for cognitive
science
so consider something like infancy you
also have things like pile
so we say there's a pile of sand and you
keep taking uh
bits of sand off at what point um how
many grains of sand are sufficient for
to
to make a pile right that's called the
sorites paradox
and the notion of the vague notion of
pile is great for active inference it's
great for generating
rapid inferences and assessments well
you know as soon as you're actually
interrogated sufficiently
the system becomes exposed the systems
flows become exposed once you're
actually subjected to all that much
scrutiny
and then same for things like book
imagine you're going to a library

and there's let's say a thousand
physical books but there's only um 800
kind of actual abstract books in the
sense that you know every library has
multiple copies of different books
so the library will have you know 10
copies of the bible uh ten copies of the
quran and so on
and let's say john goes into the library
and he reads every book in the library
and he leaves the library and because
he's fed up and all the books are
rubbish
so he just assisted decides to ban it
down in that case
uh you can say john banned every book in
the library or john read and burnt
every book in the library in which case
he banned more books than he read
right he actually bans a thousand books
but he only reads 800 books
so the phrase every book does not pick
out a fixed quantity
there's nothing in the external world
actually and exhibits a one-to-one
relation in terms of quantification it
could be 800 it could be a thousand
it depends on our perspective and that's
the crucial thing about even simple
words like book they generate these very
rich pelicans perspectives
that you can use to integrate experience
but have an unnecessary component to
them
another example is something like a city
so you can say london burned down
and was rebuilt 50 miles up the thames
and

london can still be london even though
it's physically completely changed it's
in a different location
all the londoners are dead and so on uh
london has a very complex place in a
sense
that you can decompose into organization
sense uh local
location sense uh population centers and
so on government institution
and but the single word london does not
refer to anything in the world so
in other words there's no such thing as
london that's just a kind of
convenience it's a convenient
abstraction that we use to interpret
experience and but there's nothing that
there's nothing in the world that the
word london refers to
right coherently uh you can use the word
london to refer but that's an action
it's an act of human will
to actively do that and choose to do
that so it's a choice and it's an action
again
while in the code of active entrance you
can choose to voluntarily and willfully
refer to one specific component
of your representation of a city to
refer to something in the external world
well that's a context by context case
the idea that london invariantly
refers to a particular structure is just
not
not it's not true so
by committing a more refined accurate
positioning in conceptual space
natural language syntax aids agents in

the formation of novel policies
uh to navigate and make inferences about
the environment
so cognition is an ongoing process of
dynamic interaction between an organism
and its environmental niche
uh yet notions like event are also not
predefined external entities
but are actively generated and partially
divided language system
and so again events are not things in
the world and events
are things of the mind their
construction
so consider also that simple electrical
items like city
have properties that go way beyond the
somatic complexity of other atomic
representations
so you can say the large school with
large windows next to the river
starts at 9am and has a strict
headmaster on unruly students
so there's nothing in the experimental
world that could possibly be a location
an artifact an event a social group
right surely not
absolutely that that's not an ontology
and also using these kinds of sentences
surely doesn't commit us to the belief
that there are such things in the world
it's a convenient
abstraction it's a fiction right it's
basically a picture a useful fiction
that is used in the service of active
inference and it does a very good job
it's very it's very successful and the
fact that it's so successful is

evidenced by the fact that philosophers
have only just begun to really
investigate this phenomenon this this
phenomenon is called complex gallism
and it's taken centuries of inquiry to
actually realize that some of
our most basic common nominals uh do not
have reference
to things in the world they're just
convenient fictions
so take another sentence the average man
is concerned about wage cuts
because he needs to afford insurance
does language commit to the belief that
the world is made of things like average
men and wage goods and relations of
concern
right that's not surely not something
that we're committed to
london can be as i said lonely can be
fun and polluted and burned down and
rebuilt
we will 10 miles up the river and
silvico london
uh so these nested tree structures i
mentioned earlier are widely considered
to be abstract
or even simple words exhibit
considerable abstraction
so in other words the the the the paper
i've written
with fristen and holmes focuses on
nested truth structures
and but it's also worth pointing out
that even simple ways themselves
uh exhibit a considerable degree of
abstraction
and perhaps just irrelevant here is

between richard's invitation to consider
a blind physicist who knows all the
physics right in some kind of
hypothetical
physics complete scenario so what is it
that a sighted person knows that the
blind physicist doesn't know
if this physicist knows everything
certain experience of content surely
right so what it's like to see the color
red
that's not part of the blind physicist
knowledge so therefore physics can only
capture
the causal skeleton of the word we can
at least conclude from this
that my experience of seeing the color
red simply is a property of the world
but one that we can't provide any
naturalistic account for and the reason
why i mentioned that is because i think
that may also be the case
for words like london and city and book
these abstractions are considerably
uh much more detailed much more
intricate and much more complex than
we usually give them credit for uh
our minds have managed to achieve an
analysis of the concept of number
right number theory is very rich a very
serious uh
field in mathematics about about number
theory um
but there isn't really much of an analog
in linguistics so lexical semantics
is nowhere near us nowhere near as
detailed and nowhere near as
sophisticated as number theory

okay less cosmetics is pretty much the meaning of water

is you know the set of all things that are water that's basically if you pick up a semantics textbook

semantics textbooks are pretty much just that like the meaning of the meaning of it is raining

just means uh you know it's raining right that's it just like a redescription

and so in other words linguists are very far from actually having a serious naturalistic account

for even some of the most simplest ways um

which again might be just might just simply be because of our

uh cognitive limitations right we can't construct theories for these objects

um so one of the other kind of ways to exhibit this rich politically

is by looking into the philosophy literature so in the philosophy literature

um there's something called externalism which is the position i've just been critiquing which is the idea that words can have a kind of one-to-one reference

with uh things in the external world and there was a survey conducted not too long ago

which showed that the majority of philosophers are externalists they they do believe in fact that you know the way the water refers to H_2O or whatever

so consider this famous um thought

experiment which i think
contributes to our understanding of
active inference as i'll show
in a few slides in some parallel
universe
it said that water is not made of H_2O
but rather some other substance right
xyz
so in the parallel universe planet f2
the exact same as planet f1 except water
is not made of H_2O it's xyz so the
question is can the inhabitants of this
twinner
use water to refer to the substance so
externally saying no
extenders say the meaning of water can't
be applied to the substance
and on the other hand in contrary in
contrast to the externalists there's
what's called the internalist position
which says that the meaning of words is
simply a conceptual structure
and that's it there's no there's nothing
in the external world
that these things refer to so the
internals obviously say yes of course
they can't
use it it's just a concept and so the
term water seems to be polissimous
between some more kind of common
function based sense and a more concrete
like technical sense so you can imagine
that um
let's let's say one of the examples that
numerous given is imagine that there's a
tea factory
that kind of explodes and some of the
tea leaves in the factory

get into the local water system and so
what comes out someone's tap
is chemically identical to the cup of
tea that they're making in the kitchen
and yet one of them one of the
substances is water and one of the
substances is tea
even though they're chemically identical
right chemically the same thing and you
know one of them's water
and one of them is t and the reason why
is because it doesn't set one of them
satisfies the functional based criteria
and the other one violates it indeed you
can imagine another parallel universe
so then paul petrovsky offers what he
calls paternal earth
where doppelgangers of our scientists
discover that when
um what they've all been loosely calling
mud in fact has a deep uniform structure
so obviously on planet f r planet f and
there's no uniform structure to mode
right mud can be anything
returns all that in this parallel
universe uh all of their
uh examples of substances of mud
actually exhibit a uniform structure xyz
and so the argument is that they can use
the concept mode to refer successfully
to all
physical structures of mud okay that's
good for them right they could
successfully
use the way the mode to just refer to
xyz what does it follow
from this that the inhabitants of
fraternal earth could not then travel to

our universe
and use the word mud to refer to our
chemically diverse samples
right if they if they came to a black
hole and i think the answer is no the
externalists would say yes
right the externs would say well their
meaning simply refers to
xyz and since we don't have xyz in our
universe
when these people jump in a black hole
and come to us and when they talk about
they have to be speaking of something
else
but that surely violates the actual
meaning of the term word it's conceptual
representation it's not a physical
structure so the idea that
their natural kind of forming use of mud
could not readily be extended to a
policymaker sense
doesn't really seem to be well supported
and it's just not a good description of
what language actually is or what it
cares about
language doesn't really care about the
world it's it's it cares about when i
say
you know it doesn't care about the
external world as actually is it's not
it's that's what science tries to do
science tries to achieve direct
reference to things in the world but the
language system just cares about
active inference just cares about making
sense of the work that's it
it doesn't actually care if water is
made of pastry oil right that's not

relevant

so we can use simple words like water in
wood to access multiple concepts
and then use those concepts in the sales
of active inference

in fact the trustee goes further and he
shows that using government statistics
u.s government statistics

he notes how diet coke has a high
percentage of h₂o

than stuff from my world in fact i'm
drinking uh a bottle of dr pepper

i'm not sure it actually has that
information but dr pepper has

almost definitely a greater content or
pastoral and stuff from uh the world

in your backyard and in fact that it's
bright and good so there are even more
like h₂o

and yet they're not deemed water for
reasons purely to do with intended
purposes right

so i think a cup of tea is like 99.7 or
99.5 percent h₂o

and yeah it's called a cup of tea right
it's not water it's tea

so moving even further away from this
consider that even scattered entities
forget about water let's talk about
scattered identities scatter identities
can be taken to be a single physical
object

under some conditions so imagine a
picket fence with brakes

or a colder mobile right the latter is a
thing

i'm a column mobile it's called a thing
whereas a collection of leaves on a tree

is not a thing

and unless of course these leaves are
placed for like the purposes of
decoration or art installation

and so the reason seems to be that the
mobile is created by an act of human
will

again the functional ocean is important
here so here's the question

how are these human specific notions of
function and intention coded into the
lexicon and how are they coded as part
of any genetic model

under active inference that's a really
tricky question right

and and indeed

going beyond this too between russell

and famously claim that objective is
based on spatial chemical continuity

uh but it also seems to be not

sufficient so the four legs of a dog can
be seen as a single object under many
conceivable contacts

such as if they were you know cut off
tied together and used as a doorstep but
they could still be understood by its
user as part of the dog

so abstract objects do not bear causal
relationships and they're also not
spatially temporarily located

so an object is usually understood to be
a concrete thing hence the confusion you
know when someone denied

space temporal relations so an object is
an object if we deem it so

and in addition i think it's important
to note that a psycholinguistic lens is
needed too

so when philosophers talk about externalism on internalism they often just talk about language without actually knowing anything about linguistics which is kind of like a philosopher physicist like a philosopher a philosopher of physics trying to do philosophy or physics without knowing anything about physics that's kind of strange right if you want to do philosophy of linguistics philosophy of language you should really know about linguistics so here's one particular paradox in the philosophy literature the following contrast has been called a paradox like a problem so you can say batman fights more mobsters than bruce wayne and but we also know that bruce wayne just is batman right so therefore we should be able to say batman fights more monsters than batman right but we can't say that because it sounds weird and the reason why is is due to linguistics it's not because of philosophy so that there's a constraint on discourse interpretation in language through which whenever there's two referential expressions in a clause they're default interpreted as non-identical okay and this feeds into non redundant computation which again feeds back to efficient computation and the such reference is objective so

we have two instances of batman
there's a problem so the sentence in b
forces us to search for different
reference
even though we know is the same
reference right
so this paradox and philosophy of
language is not due to
mind world relations it's just due to
linguistics it's just a psycholinguistic
phenomenon
that's all it is it's pretty simple
and again one of the problems here is
that a lot of these
uh quotes of language demand
interrogation and they're not
immediately obvious they kind of
language is very good at constructing
an illusion that we kind of become
susceptible to we like to think that
the things we talk about uh are really
really existing in the world
but in fact you know language use i like
to think it was kind of a fairy tale
you know using language is kind of more
akin to the constructing a fairy tale
than it is
science because we're just constructing
concepts and using them in a kind of
loose freaking sony
language game sets so here's another
example uh what are called escher
sentences
so there's the famous staircase painting
of M.C. Escher the endless staircases go round
and round and round
so the the visual system doesn't care
about that the visual system just sees

what it sees

if the world if that turns out to be a

physically impossible construct

that's not relevant to the visual system

we just see whatever it sees

and it's the same with language too so

language also has things

like analogous things what are called

lesser senses so if you say more people

have been to russia than i have

or in michigan and minnesota more people

found mr bush's ads negative than they

did mr carries

that's actually a meaningful sentence it

doesn't mean anything right more people

have been to russia than i have

is a meaningless sense it kind of sounds

like it makes sense

but it's completely meaningless right

because you're trying to compare a fixed

finite quantity like 50 to a simple

binary yes or no right you've ever been

to russia yes or no

and more people means five or six people

right

so it's it's syntactically legal but

it's semantically incoherent

and there's plenty of sentences like

that where it's kind of the language

system is very good at

generating the illusion of meaning or

generating the illusion

of our structure when in fact if you

interrogate it there's no

there's no meaning and again this feeds

into the idea of a efficient computation

of influence generation doesn't you know

the language system doesn't want to

interrogate too deeply it just wants to
generate an inference
right what is this person trying to say
here okay
on the other hand it also generate it
also appears into the idea of like
anti-reference
so this sentence sounds uh meaningful
but there's nothing in the external
world i can refer to so there's no real
comparisons being made here
between more people and me being uh
beating through yes or no
so in conclusion it seems that any
object is much more than its material
constitution
or its function and we can also use his
origin so thomas hobbs talked about
rivers his famous example was
a river can be maybe defined by its
origin and they can kind of scare and
diverge and and
go into different paths and maybe
converge again and but also a sense of
continuity
so john locks the personhood was that
you know a person is defined by a sense
of continuous identity
and not physical constitution and so
when a child watches a cartoon of a you
know the handsome prince getting kissed
he turns into a frog and then he turns
into uh
uh you know a human again once he's
kissed again or whatever
there's a chaos and then and something
happens the child knows that's the same
person right the child watching the tv

knows that it's the same entity it's the same person uh that the prince being turned from a human to a frog and yet again that's got enough to do with reference there's nothing that could be coherently uh existed in that sense so all these uh representations seem to inspire and in addition we also have a kind of a fifth element so we can kind of call this extra linguistic biases the shaping objective so that pertains to one example there are lots of examples but one example is default marking of object services so if you say john painted the house brown this implies that he painted the external surface brown not the internal surface because we seem to have a sense of objects as being concave objects and think of scenes at all like scenes events are kind of you know we're inside scenes or outside scenes so we seem to have a kind of visually imposed sense of what are houses so if john and mary are both stood five meters from the surface of the house but mary's inside the house and john's outside it uh john is near the house but mary's not near the house mary's inside the house even though they're both equidistant the actual physical structure of the house so the house is again a functional notion it's not just a physical object it's

also a kind of a functional based interpretation um so in other words at least these five components they all contribute to active inference they all contribute to generating structures but at least these five components are somehow encoded in language and i consider that to be uh one of the biggest mysteries how are these things encoded in the lexicon and how are these how are these networks across the brain interpreted and activated during language comprehension that's an extremely problematic issue because like i said when we talk about schools being having strict headmasters and being large and near the river and uh etc and we're using pretty much all of these concepts at once and we're doing it effortlessly and yet somehow how they're actually implemented is kind of a mystery but there is interestingly precedence in recent active imprints literature for these suggestions so in a recent paper perhaps that they'll argue that the fap is most compatible with an instrumentalist theory of mental representations in which representations are useful fictions for explanatory goals right which is exactly what i've just been saying about linguistics and this is also compatible with certain models in philosophy of language the endless perspective which assume that lester items have no one-to-one

direct reference in the external world
but are basically useful fictions
they're composites of distinct
representational domains that are used
for successful
efficient interpretation and ultimately
agent survival right
and it's also compatible with continuous
models of markov blankets
uh which have been argued to be
performed kind of neil canton and help
help launching the cancer cognition
whereby the boundaries of cognition are
diluted by the school
emphasizing the interactive constructive
nature of higher cognition
and generating intellectual actionable
concept right again the crucial concept
of
actionable concept it's a useful concept
it's something that you can you know use
in some meaningful way so the long-term
storage
of frequently generated lexical
characters and the combinatorial
rules underlying their creative
deployment in language production and
comprehension
all of this seems to allow speakers to
categorize novel sensory data
into a discrete set of object hood and
event good representations
so there are a few events that cannot
actually be passed through the
simple and unique schemes provided by
which increases the likelihood of
speakers
avoiding surprising states but more

efficiently and readily
you can pass a particular situation as
an event then
that leads to the surprise minimization
right
so another recent paper notes that the
active inference model of the brain
assumes an imperative to find the most
accurate explanation
for sensory observations that is
minimally complex which has been
recruited in barlow's exploration of
minimum redundancy
and which seems to accord with how the
language system provides
the most computationally efficient
format for solving the problem
of mapping linear sensory input right so
linear sentences
to hierarchical interpretations
so from the perspective of inference
individuals need to minimize the effort
involved in meaning making so we propose
that there is increasing evidence from
theoretical linguistics and natural
language syntax
and that this structure exhibits design
principles in keeping with this criteria
so another recent paper by christian
christian's group proposes that the
goals
of speech does documentation and all
sampling data
in a way that requires the most faster
memories degree of belief updating
in a prologue with principle so we've
basically extended these claims to the
domain of natural languages

and indeed active imprints have only one
underlying imperative to minimize
generalized
energy or uncertainty i'm much work in
psycholinguistics so things like eye
tracking
you know tracking people's eye movements
during their reading senses
uh during fully gap dependencies uh you
know long distance species and sentences
where kind of elements have to relate or
agree within features
um it shows that phrase structures are
generated predictably
in anticipation of optimistic guide in
fact it turns out that even something as
simple as adjective noun phrases
are constructed predicted so things like
red boat symbol two-word phrases
they involve rich prediction
so maybe i should just stop there for a
second to evaluate and
this has been a uh awesome learning
experience
as a non-linguist there's a few
questions
from the chat and there's also just a
few other things i wrote down but a lot
of the questions have to do with
how things happen in the brain so maybe
it's worthwhile for you to just um
share however much more you'd like to
share and then hang out
as long as you'd like to answer some
questions and you're always welcome
back so you know no worries just
whatever's comfortable today let's talk
through

and then we'll continue the discussion

cool

all right well yeah this is the final

section just to kind of wrap it all up

together so

um only a few a few a few slides more

but yeah the question of how all this

relates to the brain

is absolutely essential and so

i'll try and explain that so just to

begin with that framework

uh yeah so so far i've kind of just

outlined the kind of basic philosophy of

how language is implemented in in in a

kind of

computational theory and how it seems to

be involved in efficient computation and

so on

how it might on a cognitive level

contribute to inference generation

uh but what about actual neural

implementation

so that's that's that's the kind of the

next frontier so there's the recent

paper

by embargo and arguing that

what makes a good theory is not just

generating testable predictions

it's invoking plausible possible

mechanisms mechanisms that are plausibly

realized in nature

either in your biology or genetics or

physics right it's kind of a nice

framework the idea is that

uh theory is not just there just because

they can generate

um you know true testosterone

predictions it's

it's to generate things that are plausible and i think that's a very important point so what's your current um uh the neurobiology language involves quite reasonably testing some hypotheses um and then generating poorly available explanations for results so you'll read the results section uh which shows like let's say for example theta power increases automatically coherent sentences maybe just choose a random example and so the explanation is just kind of a redescription of the results in the discussion section so it'll be like you know we've found a hippocampal theta increases in xyz um so therefore semantically coherent sentences are indexed by a hypothetical okay it's kind of a redescription but then what we need is really a pre-existing mechanistic understanding of the possible computational properties of hippocampal theta so what is hippocampal theta in the first place right otherwise what's the point i'm looking at there's no point looking at some neural response if you don't understand the computational capacities of that neural response what is that property what is that lower level mechanism and indeed there might be there usually are multiple mechanisms multiple possible mechanisms for realizing that particular signature that

you get through
brain data and what if your if your
simple output is to simply do the
experiment
and then redescribe the results using
different rhetoric
then that's not contributing to um you
know conceptual
uh progress so a lot of these ideas are
uh outlined in the recent book of mine
the isolatory nature of language
and so i'm kind of just gonna tease them
apart but if you're interested in more
of the details you
know free to contact me and i'll send
you a copy
so one can derive some elementary
properties of linguistic computation
through
a direct line of communication from the
fpp
through to endogenous oscillatory
synchronicity and linguistic behavior
which kind of comes out of the output
so under the fpp endogenous oscillations
are the type of dynamics brain dynamics
that neurons would expect to encounter
since they have genetically encoded
beliefs that the cause of
excitatory postsynaptic potentials
follows the same pattern
so active imprints can synthesize
various and silicone neurophysical
responses via a gradient descent on free
energy
such as the mismatched negativity phase
procession theta sequences place cell
activity

uh feed the camera uh phase amplitude
and so on

um and the reason why that's important
with all these mechanisms have been
involved

and implicated in language so moving
forward with these concerns neuronal
dynamics and plasticity appear to
minimize

variational free energy under a simple
genetic model

which entails prior beliefs that
pre-synaptic inputs are generated by an
external state with a quasi-periodic
orbit

recent paper showed this so the
implication is that ensembles of neurons
make inferences about each other
while individual neurons minimize their
own free energy

so generalized synchrony kind of comes
for free

it's an amazing property of free energy
minimization desynchronization is
induced by

exogenous input explaining event related
synchronization

and structure learning images in
response to causal structure

in exogenous explaining the functional
segregation

of neuronal clusters so an external like
neuron external state with a
quasi-periodic orbit is assumed to
generate the presynaptic inputs of a
given neuron

and what's interesting for me is that
low-frequency phase synchronization

emerges directly from this assumption
and also the coupled assumption that
neural dynamics maximize variation free
energy
so the reason why that's important for
language is that
one particular implication is that
models of syntax synthetic computation
grounded in these dynamics so
yeah the the model that i have in my
book but also some other recent models
from guru grafeld and
uh savannah can be said to comply with
foundational principles of the fat
so for instance endogenous low frequency
delta delta phase trapping of syntactic
nodes
could be seen as emerging as a direct
function of
generative belief updating in accordance
with vacuum imprints supplementing the
association of delta oscillations with
the cortical computations
responsible for creating hierarchical
linguistic structures so what that means
is
whenever you get a particular phrasal
node and you have phrase boundary
you seem to get some kind of unique low
frequency response so the details can be
found in
the actual papers but there's basically
some kind of unique low frequency
signature which seems to code okay this
is a phrase here's another phrase here's
a little phrase
and that kind of goes beyond the uh
level of abstraction

uh contributed by syllables and words so
it was kind of a phrase-specific
signature and so the active inference
brainwave provides clear predictions
about the neural dynamics of language
and can help bring together research
programs that are presently pursued
independently so exploring the possible
neurobiological basis
of a core feature of language a recent
paper argues that theta gamma phase after
coupling in language
which codes syllable recognition and
also predicted coding
and physical covenants been associated
with predictive coding for a little
while now
can be brought together so this theta
gamma coupling has been applied to
silver passing
modeling theta chemical dialogue as well
which appears to form part of
belief updating fact of imprints whereby
beliefs are simultaneously updated at
high
fast levels but also low slower levels
um also
their feeds uncovering has been applied
to a constraint well it's been
assumed to be a constraint in working
memory so for instance the idea is that
the number of items you can hold in
working memory is based on the number of
gamma cycles you're going to bed
in a given theta phase and there's a
kind of trade-off between fidelity
if you want to have uh hold more items
in memory then you can indeed

increase the number of gamut items to
you know seven or eight gamma cycles
within the theta circuit about seven or
eight
but then that lowers the actual
resolution
and in fact uh it's been shown that
there's a relationship between a number
of items held
and uh uh physical causal manipulation
of theta chemically in people's
skulls using some kind of extra you know
attacks or tms
and their actual performance in working
memory tasks so here the gamma doesn't
practice because implicated in just
constructing
sets in working memory um which i think
is kind of a cool idea
so in this particular model phrase level
inferences generate words contained
within the
phrase before lower levels reset for the
next phrase right
manifested as repeater through the phase
alignment
so each transition at the higher level
is accompanied by a resetting at the
lower level status
so this is in line with the suggestion
that low frequency phase can coordinate
the bundling of lexical features indexed
by class gamma cycles
within the given description via forms
of cross-treatments equal length namely
base amplitude
and also phase phase coupler ensuring
serial readout are features alongside

the transfer of syntactic identities to language external systems so there's the kind of top-down information between uh different types of based on food coupling this is kind of elaborated in my book but the basic idea is that low frequency phases coordinate and set the identity of whatever discrete representations are being fetched and linearized and combined and chunked via these uh faster gamma cycles cross-courtedly uh prescottically just means whatever part of the brain happens to be responsible for storing the representations that you care about again language is very good at talking about extracting fetching concepts from different domains so that will yield predictions for which clausal services are being extracted and kind of spoken to by this uh low frequency phase coordination operation so theta chemical going has also formed part of recent models of scheduling and updating the list of syntactic semantic features being associated with a given chunk of linguistic stimuli with ghana cycles indexing distinct data structures being coordinated by database data structures just means linguistic features so semantic or syntactic features right

like first person number gender features
what have you
so these proposals are potentially
analogous from an euro computational
perspective
so you have the same lower level generic
algorithm newer computational algorithm
that's simply fetching discrete domain
specific representations
which also kind of feeds back into the
hanes idea i mentioned earlier
where you have these different systems
morality music
mathematics language where the
computation
seems to be analogous but the
representations are different
so the same computation operating over
different representational domains
so through specifying a process theory
that explains neuronal responses during
perception and action
neuronal dynamics have previously
encountered the gradient flow on free
energy
so that's to say any neural process can
be formulated as a minimization of the
same quantity used in approximate phase
imprints
so the brain seeks to minimize free
energy which is mathematically
equivalent to maximizing but evidence
and so this view of neural responses can
be conceived uh with respect to
hamilton's principle of least action
right so all these ideas kind of
weave together in fact recently a deep
temper model for communication

was developed based on a simulated conversation between two synthetic subjects

uh showing that certain behavioral and neurotheological correlates of communication arise under variational messenger passing in particular theta chemical right so fetal gamma coupling arose from this particular synthetic dialogue so the model of syntax i've assumed in the paper assumes that syntax sorry in this particular paper assumes that syntaxes are sequences of states or words with a terminal node at the end of every sentence so a kind of phrase boundary like a wrap-up effect right like a consolidation period and with each form of syntactic structure being limited to questions and answers in a game of 20 questions in this particular paper and but the conclusion this paper are pretty much in keeping with core assumptions in linguistics concerning the inherently constructive nature of language so elementary syntactic units which are highly robust and considered across speakers of the same language provide specific and belief structures that are used to reduce uncertainty about the world through rapid and reflective categorization of events states and objects and their relations again in compliance with the fpp uh sentential representations can be

thought of as structures designed
partially
to consolidate and appropriately frame
experiences
and to contextualize and anticipate
future experiences
so the range of possible syntactic
structures available to comprehend us
provides alternative alternate
hypotheses that afford past harmonious
explanations for sensory data
and as such they preclude overfitting so
if the convexity of the
linguistic stimuli can be efficiently
mapped to a small series of regular
syntactic formats
this contributes to the brain's goal of
restricting yourself to a number of
states as i mentioned earlier
and again by mapping syntactic
structures to conceptual systems in a
manner adhering to principles of economy
language can be seen as engaging in a
series of questions and answers with
sensory data itself
right but also non-linguistic mental
states and
only through natural language can we
generate the full complexity of
wh questions you know the questions i
mentioned before cross serial
dependencies long distance dependencies
where different elements across
different structures depend on each
other
uh fully gap dependencies and so on
which permit an expansion of what kinds
of querying

the brain connects to you over sample data

so in a word only with natural language syntax can the brain

execute these particular type of queries and generate influences

so all the ways that language centered aspects again if you recall what i meant at the beginning

or other ways that language-centered aspects of human cognition

can be motivated through conformity to the fbp and act of imprints

so things like communication and narratives all of that can actually be further be derived

from a more elementary focus on synthetic computational complexity and in fact there was a paper published a couple of days ago and by again by kristen's lab

which showed that neural dynamics under active imprints are metabolically efficient and suggest that neural representations

in biological agents may evolve by approximating steepest descent in the information space towards the point of optimal inference

and again there's a pretty no that's not a bad idea to pursue in connection to neuro-linguistics in terms of the optimal inferences afforded by not just synthetic structures but again also lexical semantics so individual words that i mentioned before

so just moving to a couple of conclusions from the slides

and i've tried to show that natural
language syntax renders meaning making
and pioneer order imprints a
computationally efficient process
and it seems we makes it work right for
what the cost etel call
as a key question for future research
for active imprints
which is how biological organisms
effectively search large policy spaces
when filing into the future so
regardless of whether
you know you're actually issuing
particular economy conditions uh x y or
z
my motivation's been kind of more
general
it's just meant to consider how the fbp
can in principle provide a novel
explanation for the prevalence of
efficiency encoded
linguistic rules and indeed other
linguists might disagree with me and
disagree with the actual framing and
maybe the other linguistic frameworks
like
maybe ray jack and up spring back with
parallel architecture that might be more
appropriate too depends on your
background assumptions so it's
specifically natural language syntax
and its capacity for sentence complexity
that allows language users to expand the
scope of their predictions
about their future positions in state
space they can think of more possible
future scenarios right on platform
recording

so we've arrived at a number of
suggested explanations for the way
language implements
the construction of hierarchical
syntactic objects namely to minimize
insanity about the causes of sensory
data
to unveil a species unique format of
external hidden states
to adhere to at least epic principle um
and in fact in some cases this involves
externalization but not always
so exploring our and disposal
empirically made demand a more mature
development of the science of
computational capacity in the brain
so i basically argue that all the ways
that language sent backwards and
permission
can be motivated for the ffb can simply
be grounded through syntax
so in other words narratives are strong
candidates for constructs adhering to
active immigrants
and but the generation of synthetic
praise structures is a necessary feature
of any narrative right you need to at
least construct a phrase to generate a
narrative
so i've reviewed how the fep that
underwrites active imprints is an
expression of the principle of induction
which is additionally a principle
implemented in models of syntax
so ultimately both the fpp and synthetic
theory are empirically and conceptually
well supported constructs
and as i've argued they share a number

of intriguing commonalities
so while the ftp has produced formal
simulation supported models of many
complex cognitive mechanisms like action
perception learning and attention
and also communication on the other hand
models of syntax
have explained grammaticality intuitions
certain poverty of stimulus issues are
how kids acquire language
and the pervasive organizational role
the hierarchy has in language so
hierarchies just seems to
pretty much organize and determine
almost all linguistic scriptures
whether language is not a is not the
only domain which exhibits economy right
and suggesting a deeper grounding of
this bias in natural law
other domains include concept learning
cause of reasoning
centrifugal learning and also as i said
before memory
so importantly at the conference has
been used to account for also creative
functions
that have to do with exploration and
novelty and the reason why that's
important is because
linguists have also long argued that the
hallmark of natural language is actually
its creative aspect
the ability to freely construct an
unbounded array of hierarchy organized
expressions
with novel interpretations so you can
say sentences that have never been said
uh linguistic creativity can be framed

in terms of adherence to physical
thermodynamic conservativity if it does
so to minimize uncertainty undeveloped
in states within an individual models of
external states
so in other words the more efficiently a
language user can internally construct
meaningful
hierarchically organized structures and
more readily they can use these
structures
to contribute to the planning and
organization of action consolidation
microsoftness and endogenous monitoring
and adaptive environment
i think it's worth recalling gregor
mendel's application of complex algebra
at the barney
at the time this was deemed by many
people to be kind of weird
and more surreal and but in fact the
same may be true
double conceptual directions um for
natural language syntax and semantics
right
unconventional approaches that are often
soon antenna
into the new normal uh only time can
tell how far these directions can
actually be pursued though
all right thank you for listening
awesome you can unshare and then
i hope we can ask a few of the questions
yeah do you want to unshare your screen
oh great okay awesome
all right i will um just jump right in
with the first question
i again as a non-linguist was like

looking up a bunch of words
learning a lot so really fascinating the
first question
was um and i think it was related to
when you said uh
that language is not just about
communication despite that being a
common conception
so potentially it's about the structure
of thought or the structure of thinking
so the question was it would be great if
elliot could define
what is his definition of thought and
what is potentially the contribution
of the intracranial language research
towards answering the hard problem
wow so you started with the easy
question okay
and yeah that's a really good question
so um i i like to think of
thoughts as um i do any other language
it's kind of like a metaphor
so and you know in in some languages and
when a human being does
a long jump they only jump so in english
they we just say they jump
in japanese they actually allow flying
so in in if a long jumper in the
japanese linguist
is at the olympics and they see someone
do high jump or long jump
they could technically say they're
flying um but it's just a metaphor it's
a
i think it's the same with thor as well
so thought is just a metaphor i don't
think we have any uh it's not a
well-defined scientific natural kind

it's kind of a
useful convenience so but you know that
said i think
the way i see it is natural language
syntax allows us to fetch
particular domain-specific
representations
from all sorts of different document
domains and then construct them into
novel interpretations
the important thing is that all of this
is outside language
so the interpretation process is an
extra linguistic process the actual only
the only
linguistic specific process is just
combining a phrase they're just
combining
items putting them together and then
shipping them off to an external
interface
for interpretation and so that's the
thought process
and i don't have any deep particular
reflections on it i think the best
the the best book about this is paul
batroski's uh conjoining meanings that
came out in 2018
um which is the idea that what language
contributes to thought
is kind of what i said here today like i
kind of
uniquely encodes a kind of functional
abstraction and whereas different
cognitive subdomains like the visual
system or the olfactory system i guess
you know the
sense of and nonsense or geometric

reasoning these all contribute distinct
sub representations of a given uh
sentence
but language seems to uniquely care
about function so abstract function
so i guess i would say that if
linguistic thought
can be defined at all it's almost
definitely going to closely approximate
human specific interests and concerns
and needs and things like that
which is kind of surprising because a
lot of people just think of language as
kind of um
you know in an unbiased way
communicating with
um the conceptual structures like
thought like language is the kind of
thought system
but i don't really see it that way i
kind of see as language does seem to
bring with it
some unique conceptual contributions uh
namely
it seems to encode these human specific
functions
um which is why i said you know if you
look at water water cannot simply be the
finest hazel
it's way too simplistic it's the the
actual meaning of water is way beyond
that
and and then with respect to
intercranial stuff
i'm not too sure like so uh
entrepreneurial research is fantastic
actually uh examining in real time
actual neural responses right so again

really down to integrity it depends on
the type of electrode that you have
it's resolution it's a listening radius
how much of cortex you can actually
sample
and there's also something called the
sparse sampling problem in in
independent research where
you have very great coverage of specific
cortical loci
but then of course each patient will
have different electrodes in different
parts of the brain
there's no patient that has electrodes
everywhere they only have electrodes
where their epilepsy is supposed to be
targeted
so for research purposes obviously it
brings limitations so i don't think
we'll ever be able to have a coherent
global whole grain in you know intra
patient understanding
what we do is we submit we combine
across all these different patients
and generate a more kind of average
brain response if that makes sense
and so in that sense um it can
contribute just
the same way any other um energy or
extra
extra cranial ehg can do it depends on
the paradigm it depends on
the analysis i think the real crucial
point here is conceptual
innovation so what we really need is we
have tons of data
and from natural language experiments we
have loads of data

but we don't have all that much uh
conceptual
uh novelty so linguists are very good at
coming up with very smart paradigms for
you know very well carefully controlled
experiments
but then when it comes to actual novel
conceptual frameworks
for how these things actually map onto
brain processes i think we need much
more of that
so i think actually um intracranial
research will not be as
uh useful as uh conceptual
theoretical um you know that changes i
think that's much more important
actually yeah well
one area of utility that you didn't even
go into at all would be
machine learning and the long time
challenge of natural language parsing
and generation
and the recent approach has been throw
the big neural network at it
with the gpt and just large-scale text
modeling
and then that reminded me of what you
said about multi-scale models how we
don't over
fit the semantics even when we hear a
ton of syllables i really really really
really really am hungry
and then um you know a computer might
just spin out a p value
it's kind of like over sampling but we
don't want to over sample semantically
so there's
gonna be really interesting space for

active inference models
okay here's the awesome absolutely okay
i'll go
to the next question um can
free energy principle and this syntactic
theory framework
help explain how and why the brain
computes inner speech
the way that it does and provide the
possibility to predict what's about to
be computed in the future so how do we
think about
externalizing speech like vocalization
versus our
inner experience of speech
are those structured are those in our
voice like what is happening
i think that's a really cool question
and well i think that i think there's
also a lot of like misconceptions about
this so i actually don't see
so so most people think that and you
know external speech
is kind of crude and you know it has
nothing to do with thoughts
but internal speech is like some kind of
angelic platonic space
and like a plate like plato's cave but
actually i think that's the opposite i
think they're both the same
so an internalized speech is basically
internalized externalized speech so when
you speak to
when you listen to yourself speaking in
your in in a monologue
you're doing it in the form of external
speech you're not doing it in a
different format it's not as if

it's not as if in a speech has a
different structure and a different
encoding
and then external speech is something
different when i think to myself
when i wake up in the morning i still
think to myself in english i think to
myself in linear sentences
right that's that's how the thought is
is externalized so it's
this is why i said about the interface
between language and conceptual systems
what's happening in inner speech
is that you are externalizing in your
own head so
that so in a switch is basically a form
of externalization okay
not all forms of externalization involve
me literally saying things
you're still externalizing it you just
externalize it to yourself
so in other words actual linguistic
thought
is is pre-conscious it's subconscious
and that's why we need linguistics
departments kind of figure out you know
what are these subconscious operations
that are happening right
so we have a bunch of subconscious
marriage operations that are happening
but we don't have direct conscious
access to those
and why should we there's no reason why
we should do right but we do have direct
access
to the externalized output in our um
kind of self-generated auditory encoding
of that structure

so um i've got i can't remember what the
question was sorry but uh
yeah that reminds me of thinking through
other minds
um a recent paper in our you know active
inference community and then also
um yeah there's just so many interesting
aspects about how you
really pointed to these domain general
attributes of language
and yes i'm rethinking
language and some of the ways that we
even communicate
on the stream because you'll see people
say i'm just not sure how to say it
it's like but are you sure how to how to
think it oh
no yeah because actually if we think
through it and then we have
feedback writing and we can it's like
stigma g we're
making these meaning marks and then
we're reinterpreting that and so
it's going to be quite interesting there
here's a third question
thanks great talk given that the
sequences are
generated in nested hierarchical
structures
where would linear externalization fit
in here
can we say that they're bound by linear
externalizations and then maybe if
that's a linguistics term
what does that mean yeah it is yeah yeah
okay great
where do linear externalizations fit
into here um

uh that's a really good question um
i'm not quite sure what the question is
is trying to ask
i kind of understand what they're saying
um
so i guess okay okay so it's typically
assumed that um
linear externalization is like the
output of this of
this detected system and that it turns
out upon better analysis
that most of the world's languages uh
differences the way that the world
languages
differ is based on more phonological
differences so in other words
the way that morphemes and sounds are
produced but the actual sciences and
syntax
is kind of fairly linear so in other
words all the universal things about
language
i kind of care early on and all the
differences in the world's languages
occur very late on so late in the
interpretation process
so when you generate a structure that
you want to say you at least need to
have the syntax first
because that's the most uniform
structure now once you've got the syntax
in place
you then go about doing you know filling
in all the sound details
but the sound details are kind of
irrelevant for language that's just
an annoyance basically right so i gave
you the example of english and basque

so english and basque have the same
underlying phrase structure
but they linearize it in different ways
so um they just happen to linearize it
based on
you know whatever arbitrary conventions
haven't do arisen in those different
cultures
but the actual interpretation is exactly
the same so i would say whatever
whatever linearization
uh whenever it happens happens late
stage and it happens as an inconvenience
to the language system
interesting and sometimes people will
point to different languages or
a word that appears unique to even a
cultural experience
but it's sort of the other side of that
coin is the
truly 99.9
that's structured functionally like that
can be translated
and so it is the exception that proves
the rule just like
when we we violate syntax sometimes to
make a point like repeating a word
and then in our internal monologue maybe
even like singing or alternate
characters it's sort of like it's a
dramatic externalization
but it's the exception that proves the
rule or it's kind of like the grand
master chess player who who violates a
principle
but that is is mastery over the syntax
and we can't let those exceptions that
prove the rule

lead us to throw out the baby with the
bathwater
or however else they say it but it it
by highlighting language as functional
and the structure of thinking rather
than like um
rhetorical only in its deployment was
just connecting
two nodes and you're really opening it
up to think about what's inside
of the cranium as well and taking
measurements from there but it's what's
happening with our thought
that's being revealed in a linear string
so so so one of the um one of my
favorite
and examples of modern literature is
magical realism
and magical realism is basically uh
put getting a blender putting lots of
different random concepts in there
i'm blending them all together and
seeing what comes out so a lot of
literary like novelty literary
conventions
uh novels poetry is novel precisely
because it just it's kind of
experimental it's saying that's what
happens when you put these two random
different concepts together let's see if
they can generate a meaningful
interpretation that
is either emotionally resonant or
conceptually intriguing
and in fact some of the most important
poetry that's been written
in the english language has violated
rules of english grammar

precisely to generate these novel interpretations it kind of it's an it's an intentional violation of some kind of linguistic rule which isn't it's like a signal to the reader that there's a reason for this it sends a message it reflects something or whatever right but yeah absolutely like this these these exceptions to the rule are extremely important to like to think about my final question is how might digital discourse and multimedia be influencing structure of language structure of thought like are we synchronizing on the functional aspects with memes or are we diverging in our narratives so how is that playing out when it's more multimedia visual video than ever less of it is spoken in red which is the linearized sort of classical language and now there's unconventional languages so where does that how's that going to work yeah so so i mean i know that the capacity and the prevalence of like uh long-term digesting of like you know long sunday times articles is kind of declining and the propensity for bite-sized chunking information is like easily digestible it involves less critical thinking judgments ideological otherwise can be made rapidly inferences can be made

and so on and that is definitely on the rise however there was a linguistics book that came out i believe around 2010 and i can't remember who it was by um it was basically just analyzing the idea that so around the turn of the century obviously mobile phones came about right and everyone started texting and using all these different so instead of saying please you would say you type plz uh instead of saying mates you would say m8 or whatever all these different abbreviations um and at the time and a lot of people in england were very concerned about this because they said well all our kids are going to be you know they're going to grow up stupid because they use all these slang deviations of english uh it turns out that there's almost no evidence that this affects intelligence like people's ability to you know be smart and use language and identify language grammatically because in fact when you think about it if you use text language and if you open a whatsapp chat and you start and you text with somebody and you use all these emojis and use all these abbreviations for words it presupposes that you actually know the correct version because in order to violate the rules it presupposes that you understand them

so fighting the rules of grammar and
fighting the rules of spelling
actually exhibits comprehension of
genuine
uh you know rules of english whatever
when i call them right
the in this book on texting again i
can't remember it
but uh so i think that's a pretty good
example of when
this panic this this moral panic in in
society over like you know text
you know it's i i the answer is i don't
know you know i have no idea
but on that sort of moral panic i guess
i thought it's like um then at some
point maybe even they forget how to
spell it
p-l-e-a-s-e you know how to spell please
and then it's kind of like
our word roots you know oh i can't
believe you don't speak greek enough to
know that the word roots of this it's
like
it's become modularized so units
that were novelties at first and
exceptions
become reified within a cultural context
a shared niche
shared narrative so that like the crying
emoji does mean this
functionally and then someone could say
i can't believe that you don't know that
it used to mean this in a different
language it's sort of like
it speaks to so many of these excellent
themes
so yeah elliot awesome

guest stream thanks for your first
appearance and we're always
looking forward to what you might want
to share with us in the future so thanks
again

totally well yeah this is an ongoing
project so i'm sure i'll have more to
share

and thank you very much for having me
thank you for the conversation i really
enjoyed it

uh thank you for the questions too so
yeah totally thank you elliot and to all
audience so see you later